Project Documentation

SaaS Product Engagement & Expansion Intelligence

1. Overview

SaaS Product Engagement & Expansion Intelligence is an end-to-end data analytics project focused on analyzing SaaS customer, subscription, operational, financial, and product-engagement data.

The project transforms raw Customer Profile and Product Operations datasets into structured business insights using Excel, MySQL/SQL, Python/Pandas, and Power BI. The final Power BI dashboard provides a centralized view of business performance through KPIs, subscription trends, customer segmentation, product engagement analysis, and business recommendations.

The dashboard is organized into four main sections:

Overview

Subscription Trends

Customer Segments

Business Insights

2. Objectives

Monitor Business Performance: Track Total Revenue, Total Customers, Total Subscriptions, Average Payment, Average Discount, and Total Tax.

Analyze Subscription Trends: Understand subscription growth, payment trends, billing-cycle performance, and feature usage.

Understand Customer Segments: Analyze customers based on segment, company size, support issues, and customer stage.

Evaluate Product Engagement: Analyze feature usage and adoption patterns across customers.

Compare Business Dimensions: Enable analysis by Region and Plan Category using interactive slicers.

Generate Business Insights: Convert analytical findings into meaningful business insights and recommendations.

Build Interactive Reporting: Provide stakeholders with an easy-to-use Power BI dashboard for business analysis.

3. Scope

In-Scope

- Customer Profile dataset analysis.
- Product Operations dataset analysis.
- Data cleaning and validation.
- Missing-value and duplicate checks.
- Data-type and categorical-value standardization.
- SQL-based querying and aggregation.
- Business KPI calculations.
- Python/Pandas exploratory data analysis.
- Subscription trend analysis.
- Customer segmentation analysis.
- Product engagement analysis.
- Power BI dashboard development.
- Interactive Region and Plan Category filtering.
- Bookmark-based dashboard navigation.
- Business insights and recommendations.

Out-of-Scope

- The completed project does not include:
- Machine-learning-based churn prediction.
- Customer health scoring.
- AI or Natural Language Query integration.
- Azure OpenAI integration.
- Real-time data streaming.
- Automated customer outreach.
- Predictive revenue forecasting.
-

4. System Architecture

The project follows a structured analytics workflow from raw data preparation to business visualization.

4.1 Data Sources

The project uses two primary CSV datasets:

Customer Profile Dataset

Product Operations Dataset

The Customer Profile dataset contains customer-related information, while the Product Operations dataset contains subscription, billing, revenue/payment, feature usage, and support-related information.

4.2 Data Preparation

Excel and Python/Pandas are used for:

- Initial data inspection.
- Missing-value analysis.
- Duplicate validation.
- Data-type correction.
- Categorical-value standardization.
- Data validation.

4.3 SQL Analysis

The prepared data is analyzed using MySQL for:

- Filtering.
- Grouping.
- Aggregation.
- Business metric calculations.
- Data validation.
- Structured business analysis.

4.4 Python Analysis

Python and Pandas are used for exploratory data analysis and validation of customer, subscription, financial, and engagement patterns.

4.5 Visualization

The analyzed data is used in Power BI to create:

- KPI cards.
- Interactive charts.
- Slicers.
- Bookmark navigation.
- Business insights.
- Recommendations.
- Architecture Flow

Customer Profile + Product Operations → Data Cleaning → SQL Analysis → Python EDA → Power BI Dashboard → Business Insights

5. Data Design

The project primarily works with two business datasets.

5.1 Customer Profile Dataset

Contains customer-level information used for customer analysis and segmentation, including:

- Customer information.
- Customer Segment.
- Company Size.
- Region.
- Customer-related characteristics.

5.2 Product Operations Dataset

Contains operational and subscription-related information used for business analysis, including:

- Subscription information.
- Plan Category.
- Billing Cycle.
- Revenue / Payment.
- Discount.
- Tax.
- Feature Usage.
- Support information.

5.3 Analytical Use

The datasets are used together to provide:

- Customer analysis.
- Subscription analysis.
- Operational analysis.
- Financial analysis.
- Product engagement analysis.
-

6. Dashboard Interface

The main user interface of the project is the Power BI SaaS Business Performance Analysis Dashboard.

The dashboard contains four analytical sections:

Overview

Provides an executive-level summary of overall SaaS business performance.

Subscription Trends

Provides detailed analysis of subscription, billing, payment, feature-usage, and support trends.

Customer Segments

Provides customer-level analysis based on customer segments, company size, support issues, and feature adoption.

Business Insights

Presents key analytical findings and corresponding business recommendations.

Interactive Controls

The dashboard includes:

- Region Slicer
- Plan Category Slicer
- Power BI Bookmark Navigation

Bookmarks allow users to switch between:

Overview → Subscription Trends → Customer Segments → Business Insights

7. Workflows

7.1 Data Cleaning Workflow

The raw datasets are inspected and prepared before analysis.

The cleaning workflow includes:

1. Import and inspect the datasets.
2. Check dataset structure and column information.
3. Identify missing values.
4. Check duplicate records.
5. Validate data types.
6. Standardize relevant categorical values.
7. Correct inconsistent formats where required.
8. Validate the cleaned data.
9. Prepare the data for SQL and Python analysis.

The purpose of this stage is to ensure that downstream calculations and dashboard visualizations are based on consistent and reliable data.

7.2 SQL Analysis Workflow

The prepared data is analyzed using MySQL and SQL.

The SQL workflow includes:

1. Loading structured data into the database.
2. Validating record information.
3. Filtering relevant records.
4. Grouping data by business dimensions.
5. Performing aggregations.
6. Calculating business-level metrics.
7. Analyzing customer and subscription information.
8. Preparing results for further analysis and reporting.

SQL serves as the structured analytical layer of the project.

7.3 Python EDA Workflow

Python and Pandas are used for exploratory analysis and additional data validation.

The workflow includes:

1. Importing the prepared dataset.
2. Checking dataset dimensions.
3. Reviewing columns and data types.
4. Performing required date conversions.
5. Performing aggregations.
6. Examining distributions.
7. Analyzing business patterns.
8. Validating customer and subscription trends.
9. Creating analytical visualizations where required.

The EDA stage helps understand the data before creating the final Power BI reporting layer.

7.4 Power BI Workflow

The final analytical data is imported into Power BI.

The dashboard development workflow includes:

- Preparing fields for visualization.
- Creating DAX measures.
- Developing KPI cards.
- Creating analytical charts.
- Adding Region and Plan Category slicers.
- Creating dashboard sections.

- Configuring bookmarks.
- Organizing visuals using the Selection pane.
- Adding business insights and recommendations.
-

8. Dashboard KPIs

The dashboard contains six major business KPIs.

Total Revenue

Measures the overall revenue represented in the analyzed data and provides a high-level view of financial performance.

Total Customers

Shows the overall customer count used to understand the customer base represented in the analysis.

Total Subscriptions

Measures subscription volume and provides visibility into overall subscription activity.

Average Payment

Shows the average payment value and helps understand customer spending patterns.

Average Discount

Measures the average discount provided and supports analysis of pricing and promotional activity.

Total Tax

Shows the total tax associated with the analyzed transactions and provides an additional financial reporting measure.

9. Dashboard Analysis

9.1 Overview

The Overview section provides a high-level summary of SaaS business performance.

Monthly Average Payment Trend

Analyzes how average payment changes over time.

Revenue by Billing Cycle

Compares revenue across available billing cycles.

Monthly Feature Usage

Tracks changes in feature usage and provides visibility into product engagement.

Customer Satisfaction by Support Team

Compares customer satisfaction across support teams to understand differences in support performance.

9.2 Subscription Trends

The Subscription Trends section provides detailed analysis of subscription and engagement behavior.

Total Subscription Growth

Shows how subscription activity changes over time and helps identify overall subscription patterns.

Average Revenue/Payment Trend

Shows changes in average revenue/payment across the analyzed period.

Total Subscriptions by Year

Provides a yearly comparison of subscription activity.

Monthly Tax Collection

Shows tax-related activity across months and provides an additional view of financial activity.

9.3 Customer Segments

The Customer Segments section focuses on customer characteristics and engagement.

Customer by Segment

Shows customer distribution across available customer segments.

Average Company Revenue by Company Size

Compares average company revenue across different company sizes.

Support Issues by Category

Analyzes support issues across available issue categories.

Feature Adoption by Customer Stage

Examines how feature adoption differs across customer stages.

10. Technology Stack

Data Preparation: Microsoft Excel.

Database: MySQL.

Querying & Analysis: SQL.

Programming: Python.

Data Analysis: Pandas and NumPy.

Analysis Environment: Jupyter Notebook.

Visualization: Power BI Desktop.

Power BI Processing: Power Query and DAX.

Version Control: Git.

Repository: GitHub.

Each technology supports a different stage of the project, from data preparation and structured analysis to visualization and documentation.

11. Setup & Project Structure

The project files are organized according to their role in the analytics workflow.

```text
SaaS-Product-Engagement-Expansion-Intelligence/
│
├── data/
│   ├── raw/
│   └── cleaned/
│
├── sql/
│   └── SQL analysis files
│
├── notebooks/
│   └── Python / Jupyter analysis
```

```
│
├── dashboard/
│   └── Power BI dashboard
│
├── docs/
│   └── Project documentation
│
└── README.md
```